



AirPod: Standalone FM Earbud

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Earlier in this series, a compact FM radio was created using a touch display. From there, the design moved forward with an FM radio integrated into an earbud—though that version had a limitation: the earbud had to be reprogrammed each time the channel needed to be changed. Now, the design journey continues with a fully standalone FM earbud that no longer relies on an external MCU for channel selection or other functions. In this version, both the MCU and user interface are built entirely into the earbud itself. Fig. 1 shows the earlier prototype, a comparison of individual components (placed next to a seed for scale), and the final surface-mounted PCB.

This is the smallest working system in the series so far. That said, it does require surface-mount soldering experience. A compact MCU and single-switch interface allow users to switch FM channels (up/down) and adjust volume. The design is compact enough to fit within an earbud, but the battery capacity remains limited, restricting extended listening time. With some reconfiguration, the earbud cavity can be modified to accommodate a larger battery.

For better reception, an external or SMD antenna can also be integrated into the system. In the current version, a simple wire coil has been used inside the earbud to act as an antenna, avoiding the need for external components while keeping the design ultra-compact.

Bill of materials

Component selection is critical, as the entire FM radio must fit inside

the earbud. Ultra-compact parts are required, with the MCU and FM

module being key. As in the earlier design, the RDA5807 FM module

BILL OF MATERIALS

Name	Designator	Footprint	Quantity	Manufacturer Part
2PIN-PAD-SMD	J1, J2	2PIN-PAD-SMD	2	SMD solder pad
HDK-100832-011	Speaker	Speaker	1	HDK-100832-011
SPST slide switch	SW1	SMD-7P, 5x1.4mm	1	SS-205-AGS7P-200
Push button	SW2	KEY-SMD_4P-L3.1-W3.1-P2.00-LS4.0	1	TS3325A 250gf 025
ATTINY412	U1	SOIC-8_L4.9-W3.9-P1.27-LS6.0-BL	1	ATTINY412-SSN
RDA5807	U2	RDA5807	1	
30-50 mAh TWS bud battery	J2	SMD solder pad	1	EEMB LP401012, D10 Kp-581013X similar battery
10 μ F capacitor	C1	Through hole, D5xL11.5mm	1	ECR1HBK100MLL050011
IC Programmer Arduino/UPDI Programmer	NA	NA	1	Arduino Uno

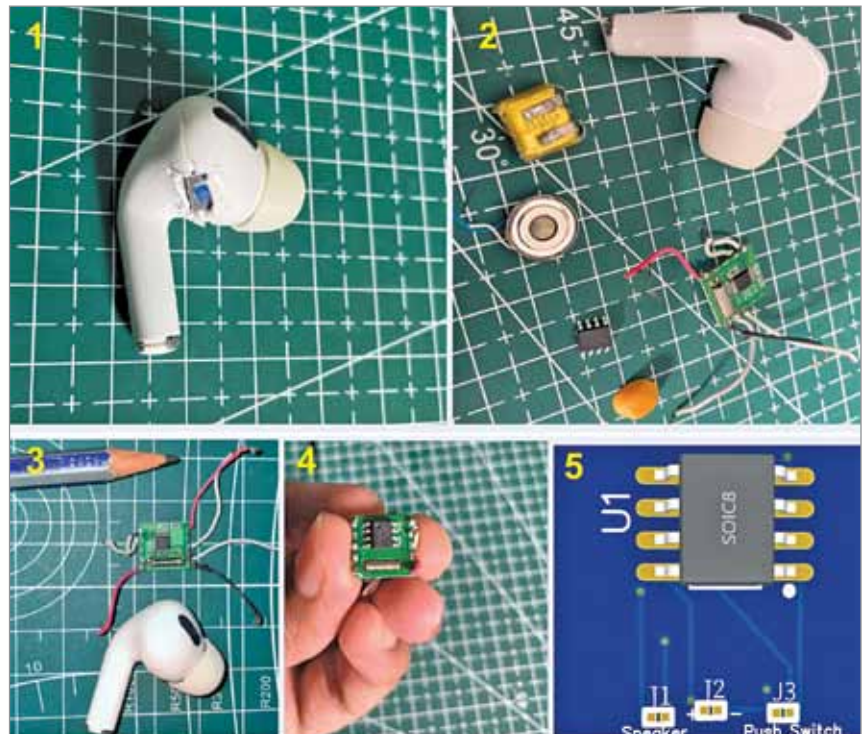


Fig. 1: Author's prototype shown in earlier issue